

Security+ 2.3 · Memory Injections

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 2.3** —
Memory Injections (2:39) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Memory injection techniques: DLL injection forces a legitimate process to load a malicious library; process hollowing replaces the memory of a running legitimate process with malicious code. Signature-based antivirus often fails to detect these because the process appears legitimate; only behavioral EDR can catch them.

High-Yield Core Concepts Tested on Exam

- DLL injection: a legitimate process loads a malicious DLL
- Process hollowing: shellcode replaces the code of a suspended legitimate process
- Signature-based AV is ineffective; behavioral EDR is required
- Both techniques allow malicious code to run under the guise of a trusted process

Exam Tips & Classic Scenarios

- 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection
- 'legitimate process loads malicious library' = DLL injection
- 'replaces memory of a running process' = process hollowing

Typical Exam Scenario: Scenario: 'malware executes inside the memory of the system's antivirus process' = DLL injection / process hollowing.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 2.3 — Memory Injections)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. An attacker enters as a regular user and becomes 'Domain Admin'. This gain of power is:

- A)** Privilege escalation
- B)** Denial of service
- C)** Social engineering
- D)** Sniffing

2. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'DLL injection: process loads attacker's DLL'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: DLL injection: process loads attacker's DLL...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

3. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'Process hollowing: shellcode replaces suspended process code'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: Process hollowing: shellcode replaces suspended process code...

4. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection

5. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: Scenario: 'malware executes inside the system antivirus process' = DLL injection / hollowing. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Scenario: 'malware executes inside the system antivirus process' = DLL injection / hollowing.
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'DLL injection: process loads attacker's DLL'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: DLL injection: process loads attacker's DLL...

7. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'Process hollowing: shellcode replaces code of a suspended process'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Process hollowing: shellcode replaces code of a suspended process...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

9. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'Scenario: malware runs inside the antivirus process' = DLL injection / hollowing.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Scenario: malware runs inside the antivirus process =...
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Memory Injections): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.3), an analyst evaluates the following technical requirement: 'DLL injection: process loads attacker's DLL'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: DLL injection: process loads attacker's DLL...

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — Privilege escalation.

✓ **Why it's correct:** Privilege escalation is the technique in which an attacker, after gaining initial access with a regular user, exploits vulnerabilities, misconfigurations, or poorly protected credentials to elevate their permission level within the system. In the context of CompTIA Security+ (SY0-701), this concept is classified as a post-exploitation tactic (TA0004 in MITRE ATT&CK) and can be vertical (from regular user to administrator) or horizontal (from one user to another with similar privileges). The statement describes exactly the transition from 'regular user' to 'Domain Admin', which is the classic example of vertical privilege escalation in Windows Active Directory environments. Therefore, the correct option is the one that identifies this movement as privilege escalation.

✗ **Why the others are wrong:**

- *Denial of service:* Denial of service (DoS) is an attack that aims to make a resource, service, or network unavailable to legitimate users, typically by overwhelming requests, exhausting resources, or exploiting flaws that cause crashes. In the described scenario, the attacker is not trying to take down the service but to gain more power within the domain. DoS does not involve privilege elevation; it affects availability, not the confidentiality or integrity of permissions. Therefore, it does not apply to the gain of power from regular user to Domain Admin.
- *Social engineering:* Social engineering is the psychological manipulation of people to reveal confidential information or perform actions that compromise security, such as clicking malicious links or providing credentials. Although social engineering can be an initial vector to gain access, it does not describe the technical process of privilege elevation within the system. The statement focuses on the gain of power after access, not the initial method of obtaining it. Therefore, social engineering is not the correct classification for the transformation from regular user to Domain Admin.
- *Sniffing:* Sniffing (or eavesdropping) is the passive interception of network traffic to capture sensitive data, such as cleartext credentials or session information. It is an information-gathering technique, not privilege elevation. Even if sniffing could lead to obtaining administrative credentials, it is not the act of escalating privileges itself. The statement describes the end result (becoming Domain Admin), not the method of traffic capture. Therefore, sniffing is not the correct answer to classify the gain of power.

2) A — Implement the corresponding solution: DLL injection: process loads attacker's DLL....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): DLL injection: process loads attacker's DLL. DLL injection is a technique where an attacker forces a legitimate process to load a malicious dynamic link library, allowing code execution within the context of that process.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

3) D — Implement the corresponding solution: Process hollowing: shellcode replaces suspended process code....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): Process hollowing: shellcode replaces suspended process code. Process hollowing is a technique where an attacker creates a legitimate process in a suspended state, hollows out its memory (replacing the original code with malicious code), and then resumes the process. This allows malicious code to run under the guise of a trusted process, evading detection. It is often used by malware to hide its execution. Mitigations include monitoring for process creation and memory modifications, and using endpoint detection and response (EDR) tools.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) D — Implement the corresponding solution: 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection. These techniques allow malware to execute within the address space of a trusted process, evading detection by security tools.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) C — Implement the corresponding solution: Scenario: 'malware executes inside the system antivirus process' = DLL injection / hollowing..

✓ **Why it is correct:** Per the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): Scenario: 'malware executes inside the system antivirus process' = DLL injection / hollowing. DLL injection is a technique in which malicious code is loaded into the address space of a legitimate running process (such as an antivirus process) so the malware executes under that process's trusted context, evading detection. Process hollowing is a related technique where a legitimate process is created in a suspended state, its memory is replaced with malicious code, and it is then resumed. Both are memory injection attacks that abuse trusted processes.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaches compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) D — Implement the corresponding solution: DLL injection: process loads attacker's DLL....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): DLL injection: process loads attacker's DLL.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) B — Implement the corresponding solution: Process hollowing: shellcode replaces code of a suspended process....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): Process hollowing: shellcode replaces code of a suspended process. Process hollowing is a technique where an attacker creates a new process in a suspended state, replaces its memory with malicious code, and then resumes it, making the malicious code appear to run under a legitimate process. The appropriate control is to implement the corresponding solution, which includes endpoint detection and response (EDR), memory protection, and behavioral monitoring.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: 'malicious code runs inside a legitimate process's memory' = process hollowing / DLL injection....

✓ Why it's correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): 'malicious code runs inside a legitimate process's memory' =

process hollowing / DLL injection. Process hollowing and DLL injection are techniques where malicious code is injected into the memory space of a legitimate process to evade detection. This matches the description exactly.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

9) C — Implement the corresponding solution: Scenario: malware runs inside the antivirus process =....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): Scenario: malware runs inside the antivirus process = DLL injection / hollowing. DLL injection and process hollowing are techniques where malicious code is injected into a legitimate process (like antivirus) to evade detection and execute with trusted privileges.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

10) D — Implement the corresponding solution: DLL injection: process loads attacker's DLL....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.3 (Memory Injections): DLL injection: process loads attacker's DLL. DLL injection is a technique where an attacker forces a legitimate process to load a malicious dynamic link library (DLL), allowing code execution within the context of that process. This can bypass security controls and enable persistence, privilege escalation, or data exfiltration. Mitigations include application whitelisting, monitoring for unusual DLL loads, and using security tools that detect injection attempts.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual

accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.